Supplementary material for "Robust leave-one-out cross-validation for high-dimensional Bayesian models"

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Abstract

Section S.1 contains details on efficient and numerically stable implementations of sampling algorithms for the mixture distribution defined in equation (9) of the main paper. Section S.2 contains additional numerical experiments to integrate the ones in Section 4 of the paper. Section S.3 contains theoretical and empirical results for the class of tempered estimators. Section S.4 contains mathematical proofs for the theoretical results stated in the paper.

S.1 Implementation details

S.1.1 Sampling from the proposed mixture with MCMC

For models with conditionally independent data as in (1) the log posterior is typically computed as the sum of log prior and log likelihood contributions as follows

$$\log p(\theta|y) = \log p(\theta) + \sum_{i=1}^{n} \log p(y_i|\theta) + const\,, \tag{S.1}$$

where *const* denotes terms that do not depend on θ. For $q_{mix}(\theta)$ defined in (9) we have the same expression plus an additional term that can be written as follows to ensure numerical

*GZ acknowledges support from the European Research Council (ERC), through *StG "PrSc-HDBayLe" grant ID 101076564*

stability

$$\log q_{mix}(\theta) = \log p(\theta) + \sum_{i=1}^{n} \log p(y_i|\theta) + LSE(\{-\log p(y_i|\theta)\}_{i=1}^{n}) + const\,, \tag{S.2}$$

where LSE denotes the usual *LogSumExp* function defined as $LSE(\mathbf{x}) = \log(\sum_{i=1}^{n} \exp(x_i))$ for $\mathbf{x} = \{x_i\}_{i=1}^{n} \in \mathbb{R}^n$. The expression in (S.2) is trivial to compute whenever the log-prior and log-likelihoods are computable and requires $\Theta(n)$ operations per evaluation, exactly as $\log p(\theta|y)$. In other words, $q_{mix}(\theta)$ can be computed up to normalizing constant whenever the original posterior $p(\theta|y)$ can. We further note that, while computing $\log q_{mix}(\theta)$ in (S.2) may appear to require roughly twice as many computations as $\log p(\theta|y)$ in (S.1), as one needs to compute both the sum and the LSE quantities, for most models the cost of computing the n likelihood terms $\{\log p(y_i|\theta)\}_{i=1}^{n}$ dominates the cost of computing their sum or the LSE function, e.g. a $\Theta(np)$ cost for the former versus a $\Theta(n)$ cost for the latter for a regression model with n data points and p covariates. Thus in such cases computing $\log q_{mix}(\theta)$ and $\log p(\theta|y)$ have roughly the same cost.

The expression in (S.2) is also trivial to differentiate, allowing to compute the gradient $\nabla \log q_{mix}(\theta)$, and is amenable to standard probabilistic programming software based automatic differentiation. For example, for the logistic regression model used in Sections S.2.4 and 4.2 the STAN code to define the posterior $p(\theta|y)$ is given by

```
data {
    int <lower=0> n;
    int <lower=0> k;
    int <lower=0, upper=1> y[n];
    matrix [n,p] X;
    real <lower=0.0> prior_scale;
}
parameters {
```

```
    vector[p] beta;
}
model{
    vector[n] means=X*beta;
    target += double_exponential_lpdf(beta | 0, prior_scale);
    target += bernoulli_logit_lpmf(y | means);
}
```

while to define the mixture distribution $q_{mix}(\theta)$ one should replace the model section with

```
model{
    vector[n] means=X*beta;
    vector[n] log_lik;
    for (index in 1:n){
        log_lik[index]= bernoulli_logit_lpmf(y[index] | means[index]);
    }
    target += double_exponential_lpdf(beta | 0, prior_scale);
    target += sum(log_lik);
    target += log_sum_exp(-log_lik);
}
```

See also the zip file in the Supplementary Material containing some of the codes to replicate the experiments of the paper for more details and examples of software implementations.

S.1.2 Efficient computation of mixture estimators

Given S samples $\{\theta_s\}_{s=1}^S$ from $q_{mix}(\theta)$, the n estimators $\{\hat{\mu}_i^{(mix)}\}_{i=1}^n$ defined in (10) can be computed at $\Theta(nS)$ total cost in a numerically stable way as follows:

(i) compute the $n \times S$ matrix of log-likelihood terms $\{\ell_{is}\}_{i,s}$, where $\ell_{is} = \log p(y_i|\theta_s)$ for $i = 1, \ldots, n$ and $s = 1, \ldots, S$;

(ii) compute the $n \times S$ matrix of log-weights $\{\tilde{w}_{is}\}_{i,s}$ defined as $\tilde{w}_{is} = \log(w_i^{(mix)}(\theta_s))$, using the equality $\tilde{w}_{is} = -\ell_{is} - \tilde{z}_s$ for $i = 1, \ldots, n$ and $s = 1, \ldots, S$, where $\tilde{z}_s = LSE(\{-\ell_{is}\}_{i=1}^n)$ for $s = 1, \ldots, S$;

(iii) compute the log-estimators exploiting the equality $\log \hat{\mu}_i^{(mix)} = \tilde{z} - LSE(\{\tilde{w}_{is}\}_{s=1}^S)$ for $i = 1, \ldots, n$ where $\tilde{z} = LSE(\{-z_s\}_{s=1}^S)$.

The above operations (i)-(iii) require $\Theta(nS)$ computational cost. In terms of memory requirements, the simplest implementation of the above operations, which creates the $n \times S$ matrices $\{\ell_{is}\}_{i,s}$ and $\{\tilde{w}_{is}\}_{i,s}$, require $\Theta(nS)$ storage, but this can be easily reduced to $\Theta(n)$ storage, if required, by storing only one column at a time.

S.2 Additional Numerical experiments

S.2.1 Decomposition of MSE in Bias and Variance components

Here we provide additional numerical illustrations on the behaviour of mean squared error (MSE) for the various estimators of $\log p(y_i|y_{-i})$ in the high-dimensional linear regression setting considered in Section 4.1.1. Relative to that section, we add to the comparison the bronze estimator and we decompose the MSE into bias squared and variance for all estimators. This allows to assess how each component contributes to the overall MSE in each case. Figure S.1 reports the results, considering an experimental setting completely analogous to the one with $\Sigma = 100/p \cdot \mathbb{I}_p$ in Section 4.1.1 of the manuscript. In particular we have $\sigma^2 = 1$, each estimator is obtained through 2×10^3 i.i.d. samples and values are averaged over 10^4 replicates (i.e. 10^4 random dataset for each n and p combination). We can see from Figure S.1 that mixture estimators have small bias squared, and their MSE is

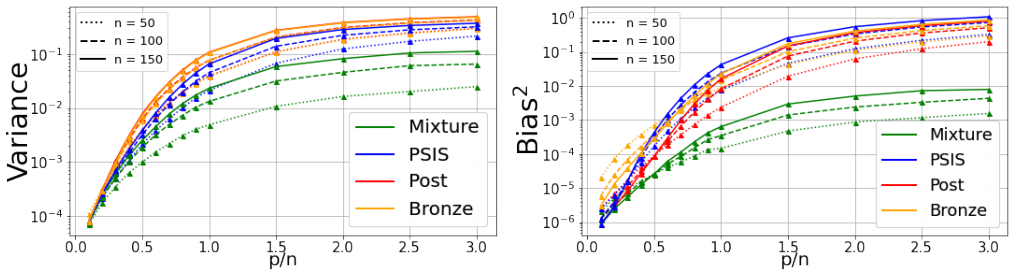


Figure S.1: Variance (left) and Bias squared (right) for the posterior, PSIS, bronze and mixture estimators of $\{\log p(y_i|y_{-i})\}_{i=1}^n$ for the high-dimensional linear regression setting of Section 4.1.1 with $\Sigma = 100/p \cdot \mathbb{I}_p$. See Section S.2.1 for more details.

indeed dominated by their variance. Both the bias and variance of other estimators under consideration are significantly larger, with their bias squared being particularly large in high-dimensional settings.

S.2.2 MSE figure for the Bladder dataset

Figure S.2 illustrate the MSE, averaging over $i = 1, \ldots, n$, for the experiments reported in Table 1 of the paper.

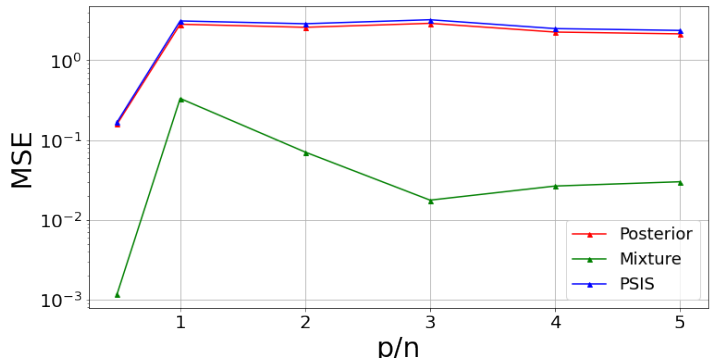
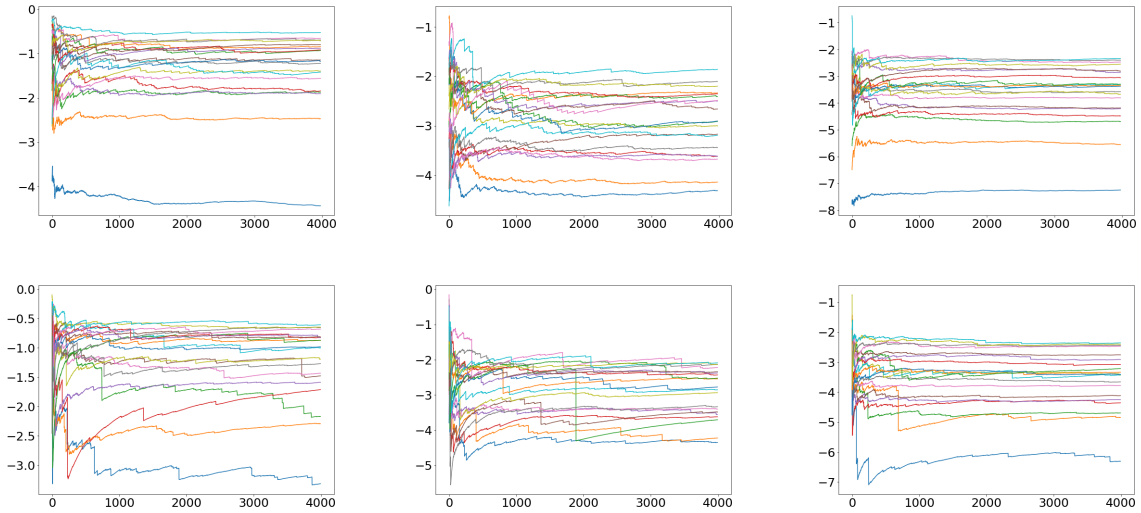


Figure S.2: Average MSE on different sub-datasets of the Bladder Cancer data. See Table 1 of the paper for more details.

S.2.3 Traceplots for high-dimensional binary regression

Figure S.3 displays the evolution of the classical and mixture estimators for the 20 data points with largest absolute value of $\log(p(y_i|y_{-i}))$, for the examples reported in Table 3 of the paper. Some classical estimators exhibit very large jumps even at high number of iterations, which is a typical pathological behaviour of estimators with infinite or excessively large variance. The mixture estimators, despite having some jumps in a few cases, display a much more stable evolution and convergence.



(a) Voice (b) Parkinson (c) Arrhythmia

Figure S.3: Evolution of the mixture (first row) and classical (second row) estimators, with number of samples on the x-axis, for three datasets (one per column). The traceplots of the estimators corresponding to the 20 data points with largest absolute value of $\log(p(y_i|y_{-i}))$ are displayed.

S.2.4 Examples from the Bayesian LOO-CV literature

In this section we consider the *Leukaemia* and *Stack Loss* datasets, which are standard example in the literature on Bayesian LOO-CV computation (Peruggia, 1997; Epifani et al.,

2008; Vehtari et al., 2017; Rischard et al., 2018). The first dataset is used to estimate the survival distribution for leukaemia patients. The response variable is survival time (from diagnosis), and the two explanatory variables are white blood cell count at diagnosis (WBC) and the outcome of a test related to white blood cell characteristics Cook and Weisberg (1982). Following previous analysis in the literature, we dichotomize survival times to indicate survival past 50 weeks, and we discard three repeated observation. The resulting dataset has $n = 30$ binary responses, $p = 3$ regressors including the intercept and is available in the Supplementary Material. We fit a Bayesian logistic regression model, meaning that each response $y_i \in \{0, 1\}$ is modelled as a Bernoulli random variables taking value 1 with success probability $(1+\exp(x_i^T\theta))^{-1}\exp(x_i^T\theta)$, where x_i is a vector of covariates. We assume independent Laplace, or double-Exponential, priors for $\theta_1, \ldots, \theta_p$ with mean parameter equal to 0 and scale parameter equal to $b = \sqrt{50/p}$, so to have prior variance for each coefficient equal to $100/p$.

This dataset is challenging for LOO-CV calculations due to the presence of a highly-influential observation, a patient with a high WBC and a survival time of more than 50 weeks, here corresponding to $i = 15$. In particular, (Epifani et al., 2008) show that for this dataset $AV_{15}^{(post)} = \infty$, while we know by Theorem 3.1.1 that $AV_{15}^{(mix)} < \infty$.

The values of $\{\mu_i\}_{i=1}^n$, where $\mu_i = p(y_i|y_{-i})$, are not available analytically, and we compute accurate approximations of them running a separate long MCMC chain to sample from $p(\theta|y_{-i})$, for each $i = 1, \ldots, n$, with 10^6 iterations and first half discarded as burn in. We treat such estimates as ground truth values, since their Monte Carlo error is negligible compared to the ones of the other estimators involved in this analysis. We then run 100 independent MCMC chains sampling from $p(\theta|y)$ and from $q_{mix}(\theta)$, of length 2×10^4 iterations each with the first half discarded as burn-in, and use the resulting samples to compute 100 i.i.d. replicates of the classical, mixture and PSIS estimators. All MCMC runs were obtain with the STAN interface in PYTHON, see e.g. `https://pystan.readthedocs.`

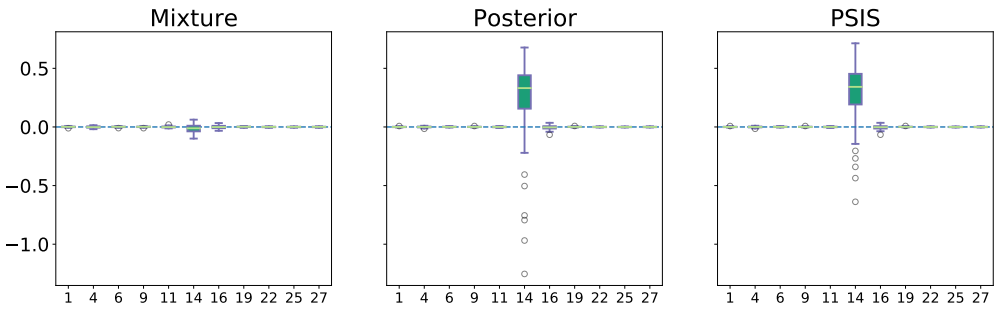


Figure S.4: Values of $\log \hat{\mu}_i - \log \mu_i$ (y-axis) across $i \in \{1, \ldots, n\}$ (x-axis) over 100 repetitions for the Leukaemia dataset ($n = 30$), where $\hat{\mu}_i$ is either $\hat{\mu}_i^{(mix)}$ (left), $\hat{\mu}_i^{(post)}$ (center) or $\hat{\mu}_i^{(psis)}$ (right). See Section S.2.4 for more details.

`io/`, using default settings, see S.1 in the Supplement for detail on how to sample from q_{mix} with STAN. No convergence or mixing issues were found using standard diagnostics.

Figure S.4 reports the results displaying, for each $i = 1, \ldots, n$, a box-plot of the differences between the log probability $\log \mu_i$ and its 100 estimates. As we can see, the classical and PSIS estimators struggle to recover the true value of $\log \mu_i$ for $i = 15$ providing highly biased estimates, which is in line with the results of Epifani et al. (2008); Vehtari et al. (2017). On the contrary, mixture estimators have drastically smaller errors and are centred around the correct values. All methods are able to accurately recover the ground truth values for the other values of i.

We now consider a second dataset previously analysed in the Bayesian LOO-CV literature, namely the *Stack Loss* dataset as in Peruggia (1997) and Vehtari et al. (2017, Section 4.3), obtaining a linear regression model with $n = 21$ observations and $p = 3$ regressors. For this example Peruggia (1997) shows $AV_i^{(post)} = \infty$ for $i = 21$. Figure S.5 displays the root mean squared error (RMSE) in estimating $\log(p(y_i|y_{-i}))$ for the problematic observation, $i = 21$, as well as a more ordinary observation, $i = 1$. We fit the model with different values of σ^2, varying them over a grid centred on the maximum marginal likelihood estimator, in order to explore sensitivity to the likelihood strength. In this example PSIS improves over

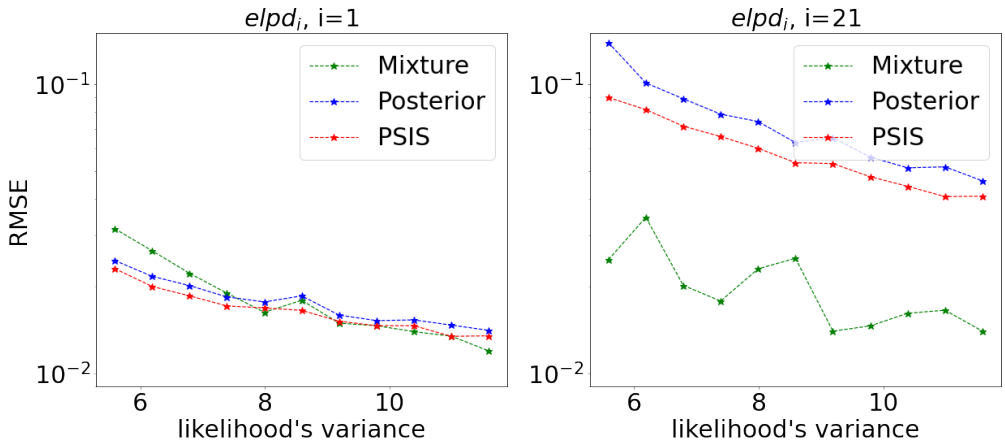


Figure S.5: Root mean squared error (RMSE) in estimating $\log(p(y_i|y_{-i}))$ for the *Stack Loss* data for $i = 1$ (left) and $i = 21$ (right). The x-axis reports the value of σ^2. See Section S.2.4 for details.

the posterior estimator for both $i = 1$ and $i = 21$. For both posterior and PSIS, the RMSE for $i = 21$ is an order of magnitude larger than the one for $i = 1$, while for the mixture they are of comparable order. As a result, mixture estimators provides a major improvement for $i = 21$, while it performs comparably for $i = 1$. This relates to the fact that mixture estimators implicitly focus more computational effort on smaller and harder to estimate values of $p(y_i|y_{-i})$ (see e.g. Remark 2 in the paper), and thus they are particularly useful for those.

S.3 Comparison with geometric tempering estimators

In this section we analyse a class of estimators based on geometric tempering, of which $\hat{\mu}_i^{(post)}$ and $\hat{\mu}_i^{(br)}$ are particular instances. First we define the class of estimators.

Definition 1. *We define the α-tempered posterior as*

$$q_{(tmp,\alpha)}(\theta) \propto p(y|\theta)^{\alpha} p(\theta) \qquad \alpha \in [0, 1] \tag{S.3}$$

and the α-tempered estimators $\{\hat{\mu}_i^{(tmp,\alpha)}\}_{i=1,\dots,n}$ as the self-normalised importance sampling estimators obtained using (S.3) *as importance distribution and $p(\theta|y_{-i})$ as target. We denote the asymptotic variance of such estimators as*

$$AV_i^{(tmp,\alpha)} = \lim_{S\to\infty} S\, var(\hat{\mu}_i^{(tmp,\alpha)}/\mu_i) \qquad\qquad i = 1,\dots,n\,. \tag{S.4}$$

Note that $q_{(1,tmp)}(\theta) = p(\theta|y)$ and $q_{(1-1/n,tmp)}(\theta) = q_{br}(\theta)$ with q_{br} as in (19) of the main paper, which implies that $\hat{\mu}_i^{(1,tmp)} \equiv \hat{\mu}_i^{(post)}$ and $\hat{\mu}_i^{(1-1/n,tmp)} \equiv \hat{\mu}_i^{(br)}$, where the latter denotes bronze estimators of (Alqallaf and Gustafson, 2001) discussed in Section 4.2.1 of the paper.

We consider the Gaussian regression model

$$\begin{aligned} y_i|\theta &\sim N(x_i^T\theta, \sigma^2) \qquad i = 1,\dots,n \\ \theta &\sim N(\theta_0, \Sigma)\,, \end{aligned} \tag{S.5}$$

although we expect the qualitative behaviour discussed here to hold similarly also in more general settings. We have the following theorem, whose proof can be found below in Section S.4.

Theorem S.3.0.1. *Under* (S.5)*, for each $i \in \{1,\dots,n\}$, we have $AV_i^{(tmp,\alpha)} < \infty$ if and only if $H_{ii}^{(\alpha)} < \frac{2}{2-\alpha}$, where*

$$H^{(\alpha)} = X\left(X^TX + \frac{\sigma^2}{2-\alpha}\Sigma^{-1}\right)^{-1}X^T. \tag{S.6}$$

Theorem S.3.0.1 suggests that lowering the tempering parameter helps in increasing the leverage value at which the estimators have infinite asymptotic variance (see e.g. (Walker and Birch, 1988) and related discussion after Theorem 3.2.1 in the paper). In particular, setting $\alpha = 0$ would produce an estimator that is guaranteed to have finite asymptotic variance since leverages are always bounded by 1. However, this would reduce to using the prior itself as importance distribution, which is well-known to be a poor choice in most commonly encountered settings (i.e. the induced asymptotic variance will be finite but very large). We illustrate this numerically in Figure S.6. We consider the model in (S.5) with

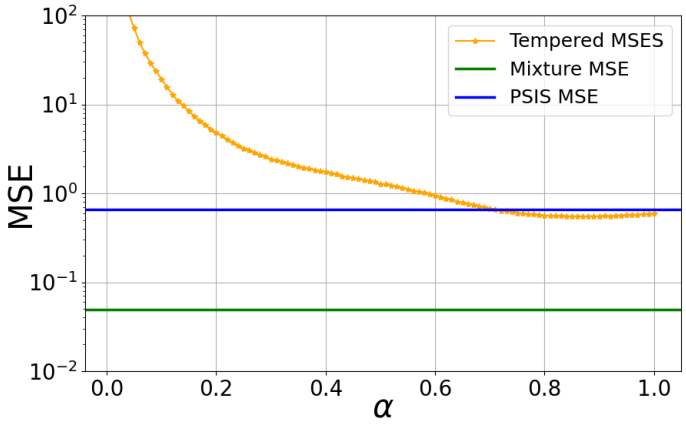


Figure S.6: Mean squared error (MSE) in estimating $\log(p(y_i|y_{-i}))$ for different tempered estimators as a function of the temperature parameter α, for a high-dimensional linear regression model with $n = p = 100$. Each estimator is computed using 2×10^3 i.i.d. samples from either $q_{(tmp,\alpha)}$ or q_{mix}. See Section S.3 for details.

$(n, p) = (100, 100)$, $\sigma^2 = 1$ and $\Sigma = 100/p \cdot \mathbb{I}_p$, and generate 10^3 synthetic datasets as in Section 4.1.1. We test the performance of different tempering estimators, each generated from 2×10^3 i.i.d. samples from $q_{(tmp,\alpha)}$. From Figure S.6 we can see that tempering only provides mild improvements over classical posterior sampling (which is the special case $\alpha = 1$) and PSIS, and starts performing very poorly when α gets small. See also Figure S.1 for more extensive results on the bias and variance of bronze estimators.

S.4 Proofs

Proof of Theorem 3.1.1. A standard application of the delta method for the derivation of the relative asymptotic variance of self-normalized importance sampling estimators, see e.g.

Liu (2001, eq.(2.7)) or Owen (2013, eq.(9.8)), applied to $\hat{\mu}_i^{(mix)}$ leads to

$$AV_i^{(mix)} = \lim_{S\to\infty} S \operatorname{var}\left(\frac{\hat{\mu}_i^{(mix)}}{\mu_i}\right) = \int \left(\frac{p(\theta|y_{-i})}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)}\right)^2 \left(\frac{p(y_i|\theta)}{\mu_i} - 1\right)^2 q_{mix}^{(\boldsymbol{\alpha})}(\theta) d\theta$$
$$= \int \frac{p(\theta|y)^2}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)} d\theta - 2\int \frac{p(\theta|y)p(\theta|y_{-i})}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)} d\theta + \int \frac{p(\theta|y_{-i})^2}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)} d\theta\,, \tag{S.7}$$

where in the last equality we re-arranged terms and used $\mu_i^{-1} p(\theta|y_{-i})p(y_i|\theta) = p(\theta|y)$. Writing $q_{mix}^{(\boldsymbol{\alpha})}(\theta) = \sum_{j=1}^n \pi_j p(\theta|y_{-j})$ with $\pi_j = Z_{\boldsymbol{\alpha}}^{-1}\alpha_j p(y_{-j})$ and upper bounding the negative terms in (S.7) by 0, we have

$$AV_i^{(mix)} \leq \int \frac{p(\theta|y)^2}{\sum_{j=1}^n \pi_j p(\theta|y_{-j})} d\theta + \int \frac{p(\theta|y_{-i})^2}{\sum_{j=1}^n \pi_j p(\theta|y_{-j})} d\theta. \tag{S.8}$$

From $\sum_{j=1}^n \pi_j p(\theta|y_{-j}) \geq \pi_i p(\theta|y_{-i})$ it follows

$$\int \frac{p(\theta|y_{-i})^2}{\sum_{j=1}^n \pi_j p(\theta|y_{-j})} d\theta \leq \int \frac{p(\theta|y_{-i})^2}{\pi_i p(\theta|y_{-i})} d\theta = \pi_i^{-1}$$

and

$$\int \frac{p(\theta|y)^2}{\sum_{j=1}^n \pi_j p(\theta|y_{-j})} d\theta \leq \pi_i^{-1} \int \frac{p(\theta|y)^2}{p(\theta|y_{-i})} d\theta = \pi_i^{-1} p(y_i|y_{-i})^{-1} \int p(y_i|\theta)p(\theta|y) d\theta\,,$$

where in the last equality we also used $p(\theta|y_{-i})^{-1}p(\theta|y) = p(y_i|y_{-i})^{-1}p(y_i|\theta)$. Combining the above with (S.8) we obtain

$$AV_i^{(mix)} \leq \pi_i^{-1}\left(1 + p(y_i|y_{-i})^{-1} \int p(y_i|\theta)p(\theta|y) d\theta\right). \tag{S.9}$$

The latter upper bound is finite by (A1) and the fact that $\alpha_i > 0$ implies $\pi_i > 0$.

Finally, (12) follows from the usual bias-variance decomposition combined with $AV_i^{(mix)} < \infty$ and the fact that $\hat{\mu}_i^{(mix)}$, being a self-normalized importance sampling estimator finite asymptotic variance, has $\mathcal{O}(S^{-1})$ bias as $S \to \infty$. The latter is a well-known fact of which we provide a proof for completeness. Recall that

$$\hat{\mu}_i^{(mix)} = \frac{A_S}{B_S}, \quad A_S = S^{-1}\sum_{s=1}^S p(y_i|\theta_s)\frac{p(\theta_s|y_{-i})}{q_{mix}^{(\boldsymbol{\alpha})}(\theta_s)}, \quad B_S = S^{-1}\sum_{s=1}^S \frac{p(\theta_s|y_{-i})}{q_{mix}^{(\boldsymbol{\alpha})}(\theta_s)}$$

where $\theta_1, \ldots, \theta_S$ are i.i.d. samples from $q_{mix}^{(\boldsymbol{\alpha})}$. It is easy to see that $\mathbb{E}[A_S] = \mu_i$ and $\mathbb{E}[B_S] = 1$. Also, $\text{var}(A_S/\mu_i) = c_1/S$ and $\text{var}(B_S) = c_2/S$, with $c_1 = \int \frac{p(\theta|y)^2}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)} d\theta < \infty$ and $c_2 = \int \frac{p(\theta|y_{-i})^2}{q_{mix}^{(\boldsymbol{\alpha})}(\theta)} d\theta < \infty$ by $AV_i^{(mix)} < \infty$ and (S.7). Thus, by the delta method, we have

$$\begin{aligned}\lim_{S\to\infty} S|\mathbb{E}[\log(\hat{\mu}_i^{(mix)}) - \log(\mu_i)]| &\leq \lim_{S\to\infty} S|\mathbb{E}[\log(A_S/\mu_i)]| + \lim_{S\to\infty} S|\mathbb{E}[\log(B_S)]| \\ &= (c_1 + c_2)/2 < \infty\,.\end{aligned}$$

The latter implies an $\mathcal{O}(S^{-1})$ bias and thus the statement in (12). □

S.4.1 Proof of Theorem 3.2.1

Lemma S.4.1.1. *For any $h \in (0,1)$, the matrix $[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T]$ is singular if and only if $H_{ii} = h$, with H as in (14). If $H_{ii} \neq h$ then*

$$[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T]^{-1} = \\ (X^TX)^{-1} + \frac{h^{-1}}{1 - h^{-1}H_{ii}} \cdot (X^TX + \sigma^2\Sigma^{-1})^{-1} x_i x_i^T (X^TX + \sigma^2\Sigma^{-1})^{-1}\,. \quad \text{(S.10)}$$

Proof. Assume first that $H_{ii} = h$, then $x_i \neq 0$ (the zero vector has leverage zero). Multiplying $[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T]$ by the non-zero vector $(X^TX + \sigma^2\Sigma^{-1})^{-1}x_i$ yields $[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T](X^TX + \sigma^2\Sigma^{-1})^{-1}x_i = x_i + h^{-1}H_{ii}x_i = 0$. Hence we have proved that in this case $[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T]$ is singular. We now verify that (S.10) is the inverse of $[X^TX + \sigma^2 \cdot \Sigma^{-1} - h^{-1}x_i x_i^T]$ when $H_{ii} \neq h$. Multiplying the two matrices we get:

$$\begin{aligned}&\mathbb{I} + \frac{h^{-1}}{1 - h^{-1}H_{ii}} x_i x_i^T (X^TX + \sigma^2 \cdot \Sigma^{-1})^{-1} - h^{-1}x_i x_i^T (X^TX + \sigma^2 \cdot \Sigma^{-1})^{-1} \\ &\qquad + \frac{h^{-2}H_{ii}}{1 - h^{-1}H_{ii}} x_i x_i^T (X^TX + \sigma^2 \cdot \Sigma^{-1})^{-1} \\ &= \mathbb{I} + x_i x_i^T (X^TX + \sigma^2 \cdot \Sigma^{-1})^{-1} \Big(\frac{h^{-1}}{1 - h^{-1}H_{ii}} + h^{-1} - \frac{h^{-2}H_{ii}}{1 - h^{-1}H_{ii}} \Big) = \mathbb{I}\,.\end{aligned}$$

□

Lemma S.4.1.2. *Let Σ be a positive definite $p \times p$ matrix, X a $n \times p$ matrix, $\sigma > 0$ and $M = X^T X - h^{-1} x_i x_i^T + \sigma^2 \Sigma^{-1}$ with $i \in \{1, \ldots, n\}$ and $h \in (0, 1)$. Then M is positive definite if and only if $H_{ii} < h$, with H as in (14).*

Proof. Assume that $[X^T X + \sigma^2 \cdot \Sigma^{-1} - h^{-1} x_i x_i^T]$ is positive definite. If $x_i = 0$, then $H_{ii} = 0 < h$. If $x_i \neq 0$, then $(X^T X + \sigma^2 \Sigma^{-1})^{-1} x_i$ is a non-zero vector and we must have, by positive definiteness, $0 < x_i^T (X^T X + \sigma^2 \Sigma^{-1})^{-1} [X^T X + \sigma^2 \cdot \Sigma^{-1} - h^{-1} x_i x_i^T] (X^T X + \sigma^2 \Sigma^{-1})^{-1} x_i = H_{ii}(1 - h^{-1} H_{ii})$. This implies that $H_{ii} < h$.

Conversely, suppose that $H_{ii} < h$. Then $h^{-1}/(1 - h^{-1} H_{ii}) > 0$, and (S.10) shows that $[X^T X + \sigma^2 \cdot \Sigma^{-1} - h^{-1} x_i x_i^T]^{-1}$ can be written as the sum of a positive definite matrix and a positive semi-definite one. As such, it is positive definite, and $[X^T X + \sigma^2 \cdot \Sigma^{-1} - h^{-1} x_i x_i^T]$ must be positive definite as well. □

Proof of Theorem 3.2.1. The relative asymptotic variance of the classical estimator $\hat{\mu}_i^{(post)}$ can be derived in analogous way to the derivation in (S.7), with the importance distribution $q_{mix}^{(\boldsymbol{\alpha})}(\theta)$ replaced by the posterior $p(\theta|y)$. After simplifications, this leads to

$$AV_i^{(post)} = \int \left(\frac{p(\theta|y_{-i})}{p(\theta|y)} \right)^2 p(\theta|y) d\theta - 1. \tag{S.11}$$

By (S.11) and (S.5) we have

$$AV_i^{(post)} + 1 = c_1 \int exp\Big\{ \frac{2(y_i - x_i^T \theta)^2}{2\sigma^2} - \frac{(y - X\theta)^T (y - X\theta)}{2\sigma^2} - \frac{(\theta - \theta_0)^T \Sigma^{-1} (\theta - \theta_0)}{2} \Big\} d\theta, \tag{S.12}$$

where c_1 is a constant independent of θ. Grouping together quadratic and linear terms in θ we obtain

$$AV_i^{(post)} = c_2 \int exp\Big\{ -\theta^T M \theta + \theta^T v \Big\} d\theta\,, \quad M = (2\sigma^2)^{-1} [X^T X - 2 x_i x_i^T + \sigma^2 \Sigma^{-1}] \tag{S.13}$$

where v is a p-dimensional vector and c_2 is a non-zero scalar, and both are independent of θ. It follows that $AV_i^{(post)}$ is finite if and only if M is positive definite. The thesis follows by Lemma S.4.1.2 with $h = 0.5$. □

S.4.2 Proof of Theorem S.3.0.1

Proof of Theorem S.3.0.1. The proof is analogous to that of Theorem 3.2.1. First, the same arguments as in (S.7) give

$$AV_i^{(tmp,\alpha)} = \int \frac{p(\theta|y)^2}{q_{(tmp,\alpha)}(\theta)}d\theta - 2\int \frac{p(\theta|y)p(\theta|y_{-i})}{q_{(tmp,\alpha)}(\theta)}d\theta + \int \frac{p(\theta|y_{-i})^2}{q_{(tmp,\alpha)}(\theta)}d\theta. \tag{S.14}$$

The first two integrals in (S.14) can be shown to be finite combining basic manipulations with the fact that $\int p(y|\theta)^\alpha p(\theta)d\theta \in (0,\infty)$ for every $\alpha \in (0,\infty)$ and that $p(y_{-i}) > 0$. Hence the finiteness of (S.14) depends on the behaviour of the last integral and we have

$$AV_i^{(tmp,\alpha)} = c_2 + \int \left(\frac{p(\theta|y_{-i})}{q_{(tmp,\alpha)}(\theta)}\right)^2 q_{(tmp,\alpha)}(\theta)d\theta = c_2 + c_1\int \frac{p(y_{-i}|\theta)^{2-\alpha}}{p(y_i|\theta)^\alpha}d\theta, \tag{S.15}$$

where c_1, c_2 are positive and finite constants that do not depend on θ. Grouping together quadratic and linear terms in θ we obtain

$$AV_i^{(tmp,\alpha)} = c_2 + \tilde{c}_1\int \exp\Big\{ -\theta^T M\theta + \theta^T v\Big\}d\theta, \tag{S.16}$$

where

$$M = \frac{2-\alpha}{2\sigma^2}\Big[X^TX - \frac{2}{2-\alpha}x_i^Tx_i + \frac{\sigma^2}{2-\alpha}\Sigma^{-1}\Big], \tag{S.17}$$

v is a p-dimensional vector and $\tilde{c}_1$ and c_2 are non-zero scalars (both independent of θ). It follows that $AV_i^{(tmp,\alpha)}$ is finite if and only if M is positive definite. The thesis follows by Lemma S.4.1.2 with $h = 1 - \alpha/2$. □

S.4.3 Proof of Proposition 3.2.2 and Theorems 3.2.3, 3.2.4 and 3.2.5

Proof of Proposition 3.2.2. Denoting $\lambda_p = \nu_p^{-2}\sigma^2$ and applying Woodbury matrix identity we have

$$\begin{aligned} H &= X(X^TX + \lambda_p \mathbb{I}_p)^{-1}X^T\,, \\ &= X(\lambda_p^{-1}\mathbb{I}_p - \lambda_p^{-1}X^T(\mathbb{I}_n + \lambda_p^{-1}XX^T)^{-1}\lambda_p^{-1}X)X^T \\ &= \lambda_p^{-1}XX^T - \lambda_p^{-1}XX^T(\mathbb{I}_n + \lambda_p^{-1}XX^T)^{-1}\lambda_p^{-1}XX^T\,. \end{aligned}$$

Since $\lim_{p\to\infty} p\lambda_p^{-1} = \sigma^{-2}c$, by Kolmogorov's criterion of SLLN and the random design assumption (A2) on X, we have $p^{-1}XX^T \to \tau^2\mathbb{I}_n$ almost surely element-wise as $p \to \infty$, see e.g. Fasano et al. (2022, Lemma 1) for a more detailed proof of the latter statement. It follows that $\lambda_p^{-1}XX^T \to \frac{c\tau^2}{\sigma^2}\mathbb{I}_n$ almost surely element-wise as $p \to \infty$ and H converges in the same way to

$$\frac{c\tau^2}{\sigma^2}\mathbb{I}_n - \frac{c\tau^2}{\sigma^2}\mathbb{I}_n(\mathbb{I}_n + \frac{c\tau^2}{\sigma^2}\mathbb{I}_n)^{-1}\frac{c\tau^2}{\sigma^2}\mathbb{I}_n = \frac{c\tau^2}{\sigma^2 + c\tau^2}\mathbb{I}_n\,,$$

which implies the desired convergence of H_{ii}. The statement about $AV_i^{(post)}$ follows by combining the above result with Theorem 3.2.1. □

Proof of Theorem 3.2.3. The statement follows from part (a) of Theorem 3.2.4, since (S.5) is a special case of (16). □

Proof of Theorem 3.2.4. First we prove part (a). Using $\pi_i^{-1} = \left(\sum_{j=1}^n p(y_j|y_{-j})^{-1}\right) p(y_i|y_{-i})$ we can re-write the upper bound in (S.9) as

$$AV_i^{(mix)} \leq \left(\sum_{j=1}^n p(y_j|y_{-j})^{-1}\right)\left(p(y_i|y_{-i}) + \int p(y_i|\theta)p(\theta|y)d\theta\right)\,. \tag{S.18}$$

By the subadditivity and submultiplicativity of lim sup, and monotonicity of $t \mapsto t^{-1}$ on

$(0, \infty)$, it follows

$$\limsup_{p\to\infty} AV_i^{(mix)} \leq \left(\sum_{j=1}^{n} (\liminf_{p\to\infty} p(y_j|y_{-j}))^{-1}\right) \left(\limsup_{p\to\infty} p(y_i|y_{-i}) + \limsup_{p\to\infty} \int p(y_i|\theta)p(\theta|y)d\theta\right). \tag{S.19}$$

We now prove that all terms on the right-hand side are finite. We have $p(y_i|y_{-i}) = p(y)/p(y_{-i})$ where, by (16),

$$p(y) \;=\; \int \prod_{j=1}^{n} g(y_j|\eta_j)p(\eta)d\eta \quad \text{and} \quad p(y_{-i}) \;=\; \int \prod_{j\neq i} g(y_j|\eta_j)p(\eta)d\eta$$

where $p(\eta) = N(\eta; 0, A_p)$ with $A_p = \nu_p^2 XX^T$ is the prior distribution on $\eta = (\eta_1, \ldots, \eta_n)$ induced by the prior on $(\theta_1, \ldots, \theta_p)$ and the linear transformation $\eta = X\theta$. As shown in the proof of Proposition 3.2.2, we have $p^{-1}XX^T \to \tau^2\mathbb{I}_n$ almost surely element-wise as $p \to \infty$, and thus also $A_p = \nu_p^2 XX^T = p\nu_p^2(p^{-1}XX^T) \to c\tau^2\mathbb{I}_n$, which implies that $p(\eta) \to N(\eta; 0, c\tau^2\mathbb{I}_n)$ almost surely as $p \to \infty$, where the convergence is point-wise in $\eta \in \mathbb{R}^n$. Also, since $A_p \to c\tau^2\mathbb{I}_n$ as $p \to \infty$, we have that, almost surely for large enough p, A_p is invertible, its determinant satisfies $|A_p| > (c\tau^2/2)^n$ and $(A_p^{-1} - (2c\tau^2)^{-1}\mathbb{I}_n)$ is positive definite. These observations imply that, almost surely, for large enough p allow we have

$$p(\eta) < (\pi c\tau^2)^{-n/2} \exp\left(-(4c\tau^2)^{-1}\|\eta\|^2\right),$$

for every $\eta \in \mathbb{R}^n$. Combining the above bound with the boundedness of the likelihood, we can apply the dominated convergence theorem and deduce that

$$p(y) \to \int_{\mathbb{R}^n} \prod_{j=1}^{n} g(y_j|\eta_j)N(\eta; 0, c\tau^2\mathbb{I}_n)d\eta = \prod_{j=1}^{n} \int_{\mathbb{R}} g(y_j|\eta_j)N(\eta_j; 0, c\tau^2)d\eta_j \in (0, \infty)$$

almost surely as $p \to \infty$. Applying the same argument to $p(y_{-i})$ we obtain

$$p(y_i|y_{-i}) = \frac{p(y)}{p(y_{-i})} \to \frac{\prod_{j=1}^{n} \int_{\mathbb{R}} g(y_j|\eta_j)N(\eta_j; 0, c\tau^2)d\eta_j}{\prod_{j\neq i} \int_{\mathbb{R}} g(y_j|\eta_j)N(\eta_j; 0, c\tau^2)d\eta_j} = \int_{\mathbb{R}} g(y_i|\eta_i)N(\eta_i; 0, c\tau^2)d\eta_i \in (0, \infty), \tag{S.20}$$

meaning that $\limsup_{p\to\infty} p(y_i|y_{-i}) < \infty$ and $(\liminf_{p\to\infty} p(y_i|y_{-i}))^{-1} < \infty$ almost surely. By the same argument we also have $(\liminf_{p\to\infty} p(y_j|y_{-j}))^{-1} < \infty$ for $j = 1, \ldots, n$. Finally,

by (16) and Bayes Theorem, we can write

$$\int p(y_i|\theta)p(\theta|y)d\theta = \frac{\int p(y_i|\theta)p(y|\theta)p(\theta)d\theta}{\int p(y|\theta)p(\theta)d\theta} = \frac{\int_{\mathbb{R}^n} g(y_i|\eta_i)^2 \prod_{j\neq i} g(y_j|\eta_j)p(\eta)d\eta}{\int_{\mathbb{R}^n} \prod_{j=1}^n g(y_j|\eta_j)p(\eta)d\eta}$$

and applying dominated convergence arguments analogous to above we obtain

$$\int p(y_i|\theta)p(\theta|y)d\theta \to \frac{\int_{\mathbb{R}} g(y_i|\eta_i)^2 N(\eta_i;0,c)d\eta_i}{\int_{\mathbb{R}} g(y_i|\eta_i)N(\eta_i;0,c)d\eta_i} \in (0,\infty)\,,$$

which implies that $\limsup_{p\to\infty} \int p(y_i|\theta)p(\theta|y)d\theta$. Combining the above bounds with (S.19) we deduce $\limsup_{p\to\infty} AV_i^{(mix)} < \infty$.

Consider now part (b) and assume $\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}\exp(-\delta\eta_i^2)d\eta_i < \infty$ for some $\delta < (2c\tau^2)^{-1}$. By (S.11)

$$AV_i^{(post)} + 1 = \int \left(\frac{p(\theta|y_{-i})}{p(\theta|y)}\right)^2 p(\theta|y)d\theta = \frac{p(y_i|y_{-i})}{p(y_{-i})}\int \frac{p(y_{-i}|\theta)^2}{p(y|\theta)}p(\theta)d\theta\,.$$

By (S.20) we have $\lim_{p\to\infty} \frac{p(y_i|y_{-i})}{p(y_{-i})} = a$ for some $a \in (0,\infty)$. Thus

$$\limsup_{p\to\infty} AV_i^{(post)} + 1 = a\limsup_{p\to\infty}\int \frac{p(y_{-i}|\theta)^2}{p(y|\theta)}p(\theta)d\theta\,.$$

By (16)

$$\int \frac{p(y_{-i}|\theta)^2}{p(y|\theta)}p(\theta)d\theta = \int_{\mathbb{R}^n} \frac{\prod_{j\neq i} g(y_j|\eta_j)}{g(y_i|\eta_i)}p(\eta)d\eta \leq \left(\prod_{j\neq i}\sup_{\eta_j} g(y_j|\eta_j)\right)\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}p(\eta_i)d\eta_i$$

with $p(\eta) = N(\eta;0,A_p)$ as above and $p(\eta_i) = N(\eta_i;0,a_p^{(i)})$ where $a_p^{(i)}$ is the i-th diagonal term of A_p. By $A_p \to c\tau^2\mathbb{I}_n$ almost surely, we have $a_p^{(i)} \to c\tau^2$ and thus $(2a_p^{(i)})^{-1} > \delta$ eventually as $p\to\infty$ since $\delta < (2c\tau^2)^{-1}$. It follows

$$\begin{aligned}\limsup_{p\to\infty}\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}p(\eta_i)d\eta_i =& (2\pi c\tau^2)^{-1/2}\limsup_{p\to\infty}\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}\exp(-(2a_p^{(i)})^{-1}\eta_i^2)d\eta_i \\ \leq& (2\pi c\tau^2)^{-1/2}\limsup_{p\to\infty}\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}\exp(-\delta\eta_i^2)d\eta_i < \infty\,.\end{aligned}$$

Combining the above inequalities we obtain $\limsup_{p\to\infty} AV_i^{(post)} < \infty$ as desired.

Finally, consider part (b) and assume $\int_{\mathbb{R}} g(y_i|\eta_i)^{-1}\exp(-\delta\eta_i^2)d\eta_i = \infty$ for some $\delta > (2c\tau^2)^{-1}$. In this case, using that $A_p \to c\tau^2\mathbb{I}_n$ as $p\to\infty$ we have

$$\begin{aligned}\limsup_{p\to\infty}\frac{\prod_{j\neq i}g(y_j|\eta_j)}{g(y_i|\eta_i)}p(\eta)d\eta =&(2\pi c\tau^2)^{-n/2}\limsup_{p\to\infty}\int_{\mathbb{R}^n}\frac{\prod_{j\neq i}g(y_j|\eta_j)}{g(y_i|\eta_i)}\exp(-\eta^T A_p\eta)d\eta\\ \geq&(2\pi c\tau^2)^{-n/2}\int_{\mathbb{R}^n}\frac{\prod_{j\neq i}g(y_j|\eta_j)}{g(y_i|\eta_i)}\exp(-\delta\|\eta\|^2)d\eta\\ =&(2\pi c\tau^2)^{-n/2}\left(\prod_{j\neq i}\int_{\mathbb{R}}g(y_j|\eta_j)\exp(-\delta\eta_j^2)d\eta_j\right)\int_{\mathbb{R}}g(y_i|\eta_i)^{-1}\exp(-\delta\eta_i^2)d\eta_i=\infty\,,\end{aligned}$$

where we used the fact that $\delta\mathbb{I}_n - A_p$ is eventually positive definite as $p\to\infty$ since $\delta > (2c\tau^2)^{-1}$. □

Proof of Theorem 3.2.5. Part (a). We first show that $AV_i^{(loo)}$ diverges as $p\to\infty$. A derivation analogous to (S.7), with the importance distribution $q_{mix}^{(\boldsymbol{\alpha})}(\theta)$ replaced by the LOO posterior $p(\theta|y_{-i})$ and some simple algebraic simplifications, leads to

$$AV_i^{(loo)} = \int\frac{p(\theta|y)^2}{p(\theta|y_{-i})}d\theta - 1 = \frac{p(y_{-i})}{p(y)^2}\int p(y_i|\theta)p(y|\theta)p(\theta)d\theta - 1\,,$$

where we also used the conditional independence assumption $p(y|\theta)=\prod_{i=1}^n p(y_i|\theta)$. Combining the above with (16) we have

$$AV_i^{(loo)}+1 = \frac{\left(\int\prod_{j\neq i}g(y_j|\eta_j)p(\eta_{-i})d\eta_{-i}\right)\left(\int h_i(\eta)p(\eta)d\eta\right)}{\left(\int\prod_{j=1}^n g(y_j|\eta_j)p(\eta)d\eta\right)^2}\,, \tag{S.21}$$

where $p(\eta_{-i})$ and $p(\eta)$ denote the prior distributions of η_{-i} and η under (16), and $h_i(\eta) = g(y_i|\eta_i)\prod_{j=1}^n g(y_j|\eta_j)$. Since $p(\eta)=N(\eta;0,A_p)$ with $A_p=\nu_p^2XX^T$ and $p(\eta_{-i})=N(\eta_{-i};0,A_p^{(i)})$ with $A_p^{(i)}=\nu_p^2X_{-i}X_{-i}^T$, we can rewrite $AV_i^{(loo)}+1$ as

$$\sqrt{2\pi p\nu_p^2\frac{|\frac{1}{p}XX^T|}{|\frac{1}{p}X_{-i}X_{-i}^T|}}\frac{\left(\int\prod_{j\neq i}g(y_j|\eta_j)K_{-i}(\eta_{-i})d\eta_{-i}\right)\left(\int h_i(\eta)K(\eta)d\eta\right)}{\left(\int\prod_{j=1}^n g(y_j|\eta_j)K(\eta)d\eta\right)^2}, \tag{S.22}$$

where $K(\eta)=\exp\left(-\eta^T(2\nu_p^2XX^T)^{-1}\eta\right)$ and $K_{-i}(\eta_{-i})=\exp\left(-\eta_{-i}^T(2\nu_p^2X_{-i}X_{-i}^T)^{-1}\eta_{-i}\right)$. We now analyze the limiting behaviour of each term in (S.22). First we have $\lim_{p\to\infty}\frac{|\frac{1}{p}XX^T|}{|\frac{1}{p}X_{-i}X_{-i}^T|}=\tau^2$ since $p^{-1}XX^T\to\tau^2\mathbb{I}_n$ and $p^{-1}X_{-i}X_{-i}^T\to\tau^2\mathbb{I}_{n-1}$ almost surely, as shown in the proof

of Proposition 3.2.2, and the determinant is a continuous function. Note that the latter convergences also imply that XX^T and $X_{-i}X_{-i}^T$ are almost surely eventually invertible as $p \to \infty$ so that K and K_{-i} are well defined. Then $K(\eta) \leq 1$ implies

$$\int \prod_{j=1}^{n} g(y_j|\eta_j) K(\eta) d\eta \leq I_y < \infty \,,$$

where $I_y = \int \prod_{j=1}^n g(y_j|\eta_j) d\eta = \prod_{j=1}^n \int g(y_j|\eta_j) d\eta_j$ is a positive and finite constant by the assumptions in part (a). Also, $K(\eta) = \exp\left(-\frac{1}{2p\nu_p^2}\eta^T(p^{-1}XX^T)^{-1}\eta\right)$ combined with $p^{-1}XX^T \to \tau^2\mathbb{I}_n$ and $p\nu_p^2 \to \infty$ implies that $K(\eta) \to \exp(0) = 1$ for every $\eta \in \mathbb{R}^n$ almost surely as $p \to \infty$, and similarly also $K_{-i}(\eta_{-i}) \to \exp(0) = 1$ for every $\eta_{-i} \in \mathbb{R}^{n-1}$. It follows by Fatou's lemma that

$$\liminf_{p\to\infty} \int \prod_{j\neq i} g(y_j|\eta_j) K_{-i}(\eta_{-i}) d\eta \geq I_{y_{-i}} \quad \text{and} \quad \liminf_{p\to\infty} \int h_i(\eta) K(\eta) d\eta_{-i} \geq I_{\tilde{y}} \,,$$

where $I_{\tilde{y}} = \int h_i(\eta) d\eta$ and $I_{y_{-i}} = \prod_{j\neq i} \int g(y_j|\eta_j) d\eta_j$ are positive and finite constants by the assumptions in part (a). Combining the above results with (S.22), the submultiplicativity of the lim inf and $p\nu_p^2 \to \infty$, we get

$$\liminf_{p\to\infty} AV_i^{(loo)} + 1 \geq \sqrt{2\pi\tau^2} \frac{I_{y_{-i}} I_{\tilde{y}}}{I_y^2} \liminf_{p\to\infty} \sqrt{p\nu_p^2} = \infty \,,$$

as desired.

We now prove that also $AV_i^{(mix)}$ diverges as $p \to \infty$ under the assumptions of part (a). By (S.7) and $\frac{p(\theta|y_{-i})}{q_{mix}(\theta)} \leq \pi_i^{-1}$ we have

$$AV_i^{(mix)} \geq \int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta - 2 \int \frac{p(\theta|y_{-i})}{q_{mix}(\theta)} p(\theta|y) d\theta \geq \int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta - 2\pi_i^{-1} \,,$$

which implies

$$\liminf_{p\to\infty} AV_i^{(mix)} \geq \liminf_{p\to\infty} \int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta - \frac{2}{\liminf_{p\to\infty} \pi_i} \,.$$

We now prove that $\int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta$ diverges with p and that $\liminf_{p\to\infty} \pi_i > 0$ for every i, thus deducing $\lim_{p\to\infty} AV_i^{(mix)} = \infty$ from the inequality above. First, by (16) we have

$$\int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta = \frac{\sum_{j=1}^n p(y_{-j})}{p(y)^2} \int \frac{\prod_{i=1}^n g(y_i|\eta_i)^2}{\sum_{k=1}^n \prod_{i\neq k} g(y_i|\eta_i)} p(\eta) d\eta = \sum_{j=1}^{n} \frac{p(y_{-j}) \int h(\eta) p(\eta) d\eta}{p(y)^2}$$

with $h(\eta) = (\sum_{k=1}^n g(y_k|\eta_k)^{-1})^{-1} \prod_{i=1}^n g(y_i|\eta_i)$. Then, using $p(y_{-j}) = \int \prod_{i\neq j} g(y_i|\eta_i) p(\eta_{-j}) d\eta_{-j}$ with $p(\eta_{-j}) = (2\pi\nu_p^2 |X_{-j}X_{-j}^T|)^{-(n-1)/2} K_{-j}(\eta_{-j})$ as defined above, we have

$$\int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta = \sum_{j=1}^n \sqrt{2\pi p \nu_p^2 \frac{|\frac{1}{p} XX^T|}{|\frac{1}{p} X_{-j}X_{-j}^T|}} \frac{\left(\int \prod_{k\neq j} g(y_k|\eta_k) K_{-j}(\eta_{-j}) d\eta_{-j}\right)\left(\int h(\eta) K(\eta) d\eta\right)}{\left(\int \prod_{j=1}^n g(y_j|\eta_j) K(\eta) d\eta\right)^2}.$$

Proceeding as done above for $AV_i^{(loo)}+1$, exploiting the almost sure point-wise convergences $K(\eta) \to 1$ and $K_{-j}(\eta_{-j}) \to 1$, one can derive

$$\liminf_{p\to\infty} \int \frac{p(\theta|y)^2}{q_{mix}(\theta)} d\theta \geq \sum_{j=1}^n \sqrt{2\pi\tau^2} \frac{I_{y_{-i}} I_{mix}}{I_y^2} \liminf_{p\to\infty} \sqrt{p\nu_p^2} = \infty\,,$$

where $I_{mix} = \int h(\eta) p(\eta) d\eta$ is a positive constant.

We now prove that $\liminf_{p\to\infty} \pi_i > 0$ for every i. From $\pi_i = \frac{p(y_{-i})}{\sum_{j=1}^n p(y_{-i})} = (1 + \sum_{j\neq i} \frac{p(y_{-j})}{p(y_{-i})})^{-1}$ it follows that

$$\liminf_{p\to\infty} \pi_i = \left(1 + \limsup_{p\to\infty} \sum_{j\neq i} \frac{p(y_{-j})}{p(y_{-i})}\right)^{-1} \geq \left(1 + \sum_{j\neq i} \limsup_{p\to\infty} \frac{p(y_{-j})}{p(y_{-i})}\right)^{-1}.$$

Then we write for every $j \neq i$

$$\frac{p(y_{-j})}{p(y_{-i})} = \left(\frac{|p^{-1}X_{-i}X_{-i}^T|}{|p^{-1}X_{-j}X_{-j}^T|}\right)^{1/2} \frac{\int \prod_{k\neq j} g(y_k|\eta_k) K_{-j}(\eta_{-j}) d\eta_{-j}}{\int \prod_{k\neq i} g(y_k|\eta_k) K_{-i}(\eta_{-i}) d\eta_{-i}},$$

which, using $p^{-1}X_{-i}X_{-i}^T \to \tau^2 \mathbb{I}_{n-1}$, $p^{-1}X_{-j}X_{-j}^T \to \tau^2 \mathbb{I}_{n-1}$, $K_{-j}(\eta_{-j}) \leq 1$ and $K_{-i}(\eta_{-i}) \to 1$, similarly to before, implies that $\limsup_{p\to\infty} \frac{p(y_{-j})}{p(y_{-i})} \leq \frac{I_{y_{-j}}}{I_{y_{-i}}} < \infty$.

Part (b). We start by proving $\limsup_{p\to\infty} AV_i^{(loo)} < \infty$. By (S.21) we can deduce

$$\limsup_{p\to\infty} AV_i^{(loo)} + 1 \leq B^2 \left(\liminf_{p\to\infty} \int \prod_{j=1}^n g(y_j|\eta_j) p(\eta) d\eta\right)^{-2} \tag{S.23}$$

where $B = \sup_{\eta\in\mathbb{R}^n} \prod_{i=1}^n p(y_i|\eta_i)$ is a finite constant by the assumption of upper bounded likelihood. Since $p(\eta) = N(\eta; 0, \nu_p^2 XX^T) \to 0$ almost surely for every $\eta \in \mathbb{R}^n$ as $p \to \infty$, it is convenient to define the change of variables $\gamma = (p\nu_p^2)^{-1/2}\eta$ and re-write the integral above as

$$\int_{\mathbb{R}^n} \prod_{j=1}^n g(y_j|\eta_j) N(\eta; 0, \nu_p^2 XX^T) d\eta = \int_{\mathbb{R}^n} \prod_{j=1}^n g\left(y_j|\sqrt{p\nu_p^2}\gamma_j\right) N(\gamma; 0, p^{-1}XX^T) d\gamma. \tag{S.24}$$

Defining $a_i = \lim_{\eta_i \to -\infty} g(y_i|\eta_i)$ and $b_i = \lim_{\eta_i \to \infty} g(y_i|\eta_i)$, we have $\lim_{p\to\infty} g\left(y_j|\sqrt{p\nu_p^2}\gamma_j\right) = (a_i(1-\mathrm{sgn}(\gamma_i))+b_i\,\mathrm{sgn}(\gamma_i))$ for every i and every $\gamma_i \neq 0 \in \mathbb{R}$ and $\lim_{p\to\infty} N(\gamma; 0, p^{-1}XX^T) = N(\gamma; 0, \tau^2\mathbb{I}_n)$ for every $\gamma \in \mathbb{R}^n$ almost surely as $p \to \infty$. Thus, by Fatou's lemma we have

$$\liminf_{p\to\infty} \int_{\mathbb{R}^n} \prod_{j=1}^{n} g\left(y_j|\sqrt{p\nu_p^2}\gamma_j\right) N(\gamma; 0, p^{-1}XX^T)d\gamma \geq$$

$$\int_{\mathbb{R}^n} \prod_{j=1}^{n} (a_i(1-\mathrm{sgn}(\gamma_i)) + b_i\,\mathrm{sgn}(\gamma_i))N(\gamma; 0, \tau^2\mathbb{I}_n)d\gamma = \prod_{j=1}^{n}\left(\frac{a_i}{2}+\frac{b_i}{2}\right) > 0\,. \tag{S.25}$$

The latter product is a positive constant by the assumption $a_i+b_i > 0$ for any i. Combining (S.25) and (S.23) we obtain $\limsup_{p\to\infty} AV_i^{(loo)} < \infty$ as desired.

We now prove $\limsup_{p\to\infty} AV_i^{(mix)} < \infty$. Equation (S.25) states that $\liminf_{p\to\infty} p(y) > 0$. An analogous derivation can be used to prove that $\liminf_{p\to\infty} p(y_{-j}) > 0$ for every $j = 1, \ldots, n$. Combining the latter with $\limsup_{p\to\infty} p(y_{-j}) \leq B_{-j} < \infty$ for every $j = 1, \ldots, n$, with $B_{-j} = \sup_{\eta\in\mathbb{R}^n} \prod_{i\neq j} p(y_i|\eta_i) < \infty$, we obtain that $\liminf_{p\to\infty} \pi_i \geq \frac{\liminf_{p\to\infty} p(y_{-i})}{\sum_{j=1}^{n} \limsup_{p\to\infty} p(y_{-i})} > 0$. One can then deduce

$$\limsup_{p\to\infty} AV_i^{(mix)} < (\limsup_{p\to\infty} \pi_i^{-1})(\limsup_{p\to\infty} AV_i^{(loo)}) < \infty$$

as desired.

To conclude, we prove $\lim_{p\to\infty} AV_i^{(post)} = \infty$. By (16) and (S.11)

$$AV_i^{(post)} + 1 = \frac{p(y)}{p(y_{-i})^2} \int_{\mathbb{R}^n} \frac{\prod_{k\neq i} g(y_k|\eta_k)}{g(y_i|\eta_i)} p(\eta)d\eta.$$

Thus

$$\liminf_{p\to\infty} AV_i^{(post)} + 1 \geq \frac{\liminf_{p\to\infty} p(y)}{B_{-i}^2} \liminf_{p\to\infty} \int_{\mathbb{R}^n} \frac{\prod_{k\neq i} g(y_k|\eta_k)}{g(y_i|\eta_i)} p(\eta)d\eta,$$

where $B_{-i} < \infty$ and $\liminf_{p\to\infty} p(y) > 0$ as shown above. Using the same change of variable of (S.24) and proceeding as in (S.25) we obtain

$$\liminf_{p\to\infty} \int_{\mathbb{R}^n} \frac{\prod_{k\neq i} g(y_k|\eta_k)}{g(y_i|\eta_i)} p(\eta)d\eta \geq \left(\frac{1}{2a_i}+\frac{1}{2b_i}\right) \prod_{j\neq i}\left(\frac{a_j}{2}+\frac{b_j}{2}\right) = \infty$$

where the latter equality follows from the assumptions that $a_i b_i = 0$ and $(a_i + b_i) \in (0, \infty)$. It follows that $\liminf_{p\to\infty} AV_i^{(post)} = \infty$ almost surely, and thus also $\lim_{p\to\infty} AV_i^{(post)} = \infty$ almost surely as desired. □

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